

This page is from yesterday. Only highlighted entries in the table are new

Recall $r \times c$ matrix A defines a map $f = f_A$
from $\mathbb{R}^c$ to $\mathbb{R}^r$ by eqⁿ $f(x) = Ax$

To solve $Ax=b$ i.e. $f(x)=b$, row reduce $A|b$ to
its reduced row echelon form RREF. Two cases

- Either system is inconsistent $\Leftrightarrow$ pivot in last col of $\text{RREF}(A|b)$
- OR $\left\{ \begin{array}{l} \text{the "dim" of solution space} = \# \text{ of non-pivot columns in} \\ \text{RREF}(A) = \# \text{ free variables} \end{array} \right.$
and then (general solution to $Ax=b$)
 $= (\text{a fixed sol}^n \text{ to } Ax=b) + (\text{gen sol}^n \text{ to } Ax=0)$

The following are equivalent TFAE / TFAE

- | | |
|---|---|
| ① f_A is injective (i.e. 1 to 1) | ①' f_A is surjective (i.e. onto) |
| ② $Ax=0$ has only the 0 sol ⁿ | ②' $Ax=b$ is consistent for each $b \in \mathbb{R}^r$ |
| ③ Every col of $\text{RREF}(A)$ has a pivot | ③' Every row in $\text{RREF}(A)$ has a pivot |
| ④ Col vectors of A are linearly indep | ④' Col vectors of A span $\mathbb{R}^r$ |
| ⑤ row vectors of A ??? exercise | ⑤' row vectors of A ??? exercise |

Remarks

- | | |
|---|------------------|
| ① IF f_A is injective then $c \leq r$ | } Converse false |
| IF f_A is surjective then $r \leq c$ | |
| IF f_A is bijective then $r = c$ | |

Recall A **vector space** (over $\mathbb{R}$) is a set V with two operations $V \times V \xrightarrow{+} V$ such that $\mathbb{R} \times V \xrightarrow{\cdot} V$

① V is an abelian group under $+$ $\left\{ \begin{array}{l} u+v = v+u, u+(v+w) = (u+v)+w, \\ \text{There is } \bar{0} \in V \text{ s.t. } v+\bar{0} = v \text{ for all } v \in V \\ \text{For any } v \in V \text{ there is } w \text{ in } V \text{ such that } v+w = \bar{0} \end{array} \right.$

② $1 \cdot v = v, (a+b) \cdot v = (a \cdot v) + (b \cdot v), a \cdot (v+w) = (a \cdot v) + (a \cdot w), (ab) \cdot v = a \cdot (b \cdot v)$

Fun and games (i) The $\bar{0}$ in V is unique. (If z is another such element then $\bar{0} + z = z = \bar{0}$)

(ii) For v , the w such that $v+w = \bar{0}$ is unique (and is denoted by $-v$). If w_1 and w_2 are two such then consider $w_2 + (v + w_1) = w_2 + \bar{0} = w_2$
 $(w_2 + v) + w_1 = \bar{0} + w_1 = w_1$

(iv) $(-1) \cdot v = -v \quad (1 + (-1)) \cdot v = 0 \cdot v = \bar{0} \xrightarrow{\text{by (iii) below}} = 1 \cdot v + (-1) \cdot v = v + (-1) \cdot v$

(iii) $0 \cdot v = \bar{0} \quad 0 \cdot v = (0+0) \cdot v$
 $0 \cdot v = 0 \cdot v + 0 \cdot v$ Add $-(0 \cdot v)$ to both sides
 $\bar{0} = 0 \cdot v$

Example of vector spaces so far: $\mathbb{R}^n$, more soon!

Defⁿ A subspace W of a vector space V is a subset of V such that W itself is a vector space under the same operations

nonempty

Useful remark: Suppose V is a v.space. A nonempty subset W is a subspace $\Leftrightarrow$ " W is closed under $+$ & $\cdot$."

i.e. for $w_1, w_2 \in W$ & $c \in \mathbb{R}$
 $w_1 + w_2 \in W$, $c \cdot w_1 \in W$

$\Rightarrow$ is immediate

$\Leftarrow$ is also true b/c $\begin{cases} \textcircled{1} \text{ \& \textcircled{2} are inherited from their} \\ \text{validity in } V. \end{cases}$

Examples of subspaces of $\mathbb{R}^n$

• $\{0\}$ and $\mathbb{R}^n$ are subspaces of $\mathbb{R}^n$

• Subspaces of $\mathbb{R}^2$ (other than 0 and $\mathbb{R}^2$).

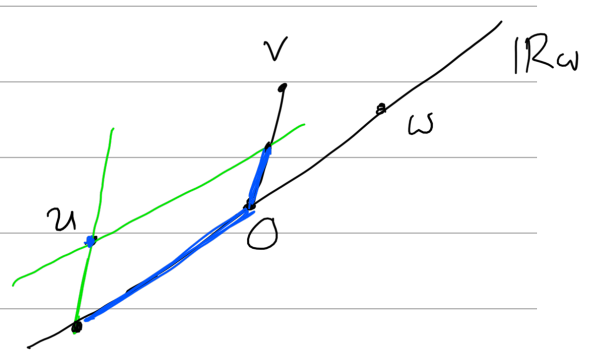
Let W be such a subspace. Take nonzero $w \in W$.

$\Rightarrow \mathbb{R}w \subset W$ ($\mathbb{R}w$ = line through origin containing w)

If $v \in W \setminus \mathbb{R}w$, then $W = \mathbb{R}^2$

Pf. Let $u \in \mathbb{R}^2$. Then I claim

$u = p v + q w$ for some $p, q \in \mathbb{R}$



i.e.
$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = p \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + q \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} p v_1 + q w_1 \\ p v_2 + q w_2 \end{bmatrix} = \underbrace{\begin{bmatrix} v_1 & w_1 \\ v_2 & w_2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} p \\ q \end{bmatrix}}_x$$

This system can be solved for $\begin{bmatrix} p \\ q \end{bmatrix}$.

i.e. we are claiming $Ax = b$ is consistent for any $b = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

This is true b/c $v_1 w_2 \neq v_2 w_1 \iff v$ and w are not collinear

- Do the same for $\mathbb{R}^3$: classify all subspaces
(Ans: 0 , line thro 0 , plane thro 0 , $\mathbb{R}^3$)

Recall matrix A $\rightsquigarrow$ map $f_A: \mathbb{R}^c \rightarrow \mathbb{R}^r$
 $x \mapsto Ax$

We have two associated subspaces, one of $\mathbb{R}^c$, one of $\mathbb{R}^r$

① $\text{Nul}(A) = \text{Nul}(f_A) = \ker(f_A) = \{x \mid Ax = 0\}$
 is a subspace of $\mathbb{R}^c$

② Range of f_A i.e. $\{Ax \mid x \in \mathbb{R}^c\} \subset \mathbb{R}^r$

$$\left. \begin{array}{l} \text{If } b_1, b_2 \in \text{range} \\ b_1 = Ax_1 \\ b_2 = Ax_2 \end{array} \right\} \Rightarrow b_1 + b_2 = Ax_1 + Ax_2 = A(x_1 + x_2) \in \text{range}$$

$$\text{And } cb_1 = cAx_1 = A(cx_1) \in \text{range}$$

Defⁿ (in any vector space V)

Let $S = \{v_1, v_2, \dots, v_n\}$ be a ~~list~~^{set} of vectors in V

- A linear combination of $v_1, \dots, v_n$ is any vector of form $x_1 v_1 + x_2 v_2 + \dots + x_n v_n$ where $x_i \in \mathbb{R}$

Span of $v_1, v_2, \dots, v_n$ = set of all linear combinations of $v_1, \dots, v_n$
 $= \left\{ \sum_{i=1}^n c_i v_i \mid c_i \in \mathbb{R} \right\}$

• Observe ① $\text{Span}(S)$ is a subspace of V

$$\left(\sum c_i v_i\right) + \left(\sum d_i v_i\right) = \sum (c_i + d_i) v_i$$
$$a\left(\sum c_i v_i\right) = \sum ac_i v_i$$

② $\text{Span}(S)$ is the smallest subspace of V containing S .

③ For a matrix A , $\text{range}(f_A) = \text{Span}(\text{column vectors of } A)$

$$\{x_1 A_1 + \dots + x_n A_n = Ax \mid x \in \mathbb{R}^n\}$$

where $A_1, \dots, A_n$ are columns of A

So f_A is surjective $\Leftrightarrow \text{span}(\text{cols of } A) = \mathbb{R}^r$
i.e. cols of A span $\mathbb{R}^r$

What does injectivity of f_A mean for cols of A ?

$$\Updownarrow$$

$$Ax = 0 \Rightarrow x = 0$$

$$\Updownarrow$$

$$x_1 A_1 + \dots + x_n A_n = 0 \Rightarrow \text{each } x_i = 0$$

i.e. "col vectors of A are lin indep"

IMPORTANT

↳ Defⁿ $v_1, v_2, \dots, v_n$ are said to be linearly

• independent if $\sum x_i v_i = 0 \Rightarrow \text{each } x_i = 0$

• dependent if we can find x_i , not all 0 st $\sum x_i v_i = 0$

Note: $v_1, \dots, v_n$ are lin dependent $\Leftrightarrow$ one of them can be written as a lin combination of others.
($n > 1$)